2.2.2 Transport systems in mammals and plants

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the importance of the closed double circulatory system	To include reference to blood pressure in systemic and pulmonary systems.
(b)	**(i)** the structure and functions of arteries, arterioles, capillaries, venules and veins	HSW4, HSW5, HSW6
	(ii) transverse sections of arteries, veins and capillaries as observed using a light microscope	Prepared slides or photomicrographs may be examined to show the structural differences between these vessels.
(c)	the formation and importance of tissue fluid	To include references to HP (hydrostatic pressure) and OP (oncotic pressure or colloidal osmotic pressure).
(d)	**(i)** the use of a sphygmomanometer to measure systolic and diastolic blood pressure	To include reference to both manual and electronic measuring.
	(ii) comparisons of blood pressure readings	HSW3, HSW4, HSW5, HSW6
(e)	the interpretation of systolic and diastolic blood pressure measurements	To include hypertension and hypotension and their possible consequences.

(f)	**(i)** transport systems in multicellular plants	To include the structure and functions of vascular tissues in roots, stems and leaves and an outline of the mechanisms of bulk transport (cohesion-tension theory in xylem and mass flow in phloem).
	(ii) the observation, drawing and annotation of sections of vascular plant tissues as seen using a light microscope.	**PAG1, PAG2, PAG5**